National Education Society®

JNN College of Engineering, Shivamogga

Dept. of Information Science and Engineering

Mini Project Progress Phase-1 presentation on

"Crop Recommendation Using Machine Learning "

Batch : 21

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1. Introduction

- Agriculture is the backbone of many economies , as we all know agriculture is a non-technical industry where technology can help with growth and improvement.
- Agricultural technology needs to be implemented quickly and in a growing number of places. This industry has employed almost 60% of the population of our country. On the other hand, it accounts for only 18 percent of India's annual GDP.
- The disparity arises from A lack of research and innovative technology .The proper selection of crops is crucial for maximizing yield and sustainability.
- However, farmers often face challenges in determining the most suitable crops for their land due to varying factors such as soil type, weather conditions, and market demand.
- Traditional methods of crop selection rely heavily on personal experience, which can lead to inconsistent results and inefficiencies.

1. Introduction

- Farmers using traditional methods for identifying which crops to grow often end up growing inefficient crops , while using technologies can help in growing much more efficient crops .
- In recent years, advancements in machine learning have opened new opportunities to revolutionize agriculture.
- A crop recommendation system powered by machine learning analyzes vast amounts of data, including soil properties, climate, and historical crop performance, to suggest the most appropriate crops for a given region.
- This not only improves crop productivity but also promotes resource-efficient farming, helping farmers make informed decisions.
- Our main goal of this project is to help farmers grow better crops by using technology.

2. Problem Statement and Description

Problem Statement:

Identifying of crops to grow on particular area based on soils pH and temperature using machine learning algorithm.

Problem Description:

In todays growing world where agriculture is one of the major developing sector of economy identifying which crop to grow has been a main problem for farmers. This projects aims to present a solution to the farmers using current advanced technology like machine learning. The goal of this project is to develop a model or framework that can recommend suitable crops based on input parameters like pH and temperature values of soil.

3. Objectives

Following are the objectives of the project undertaken :

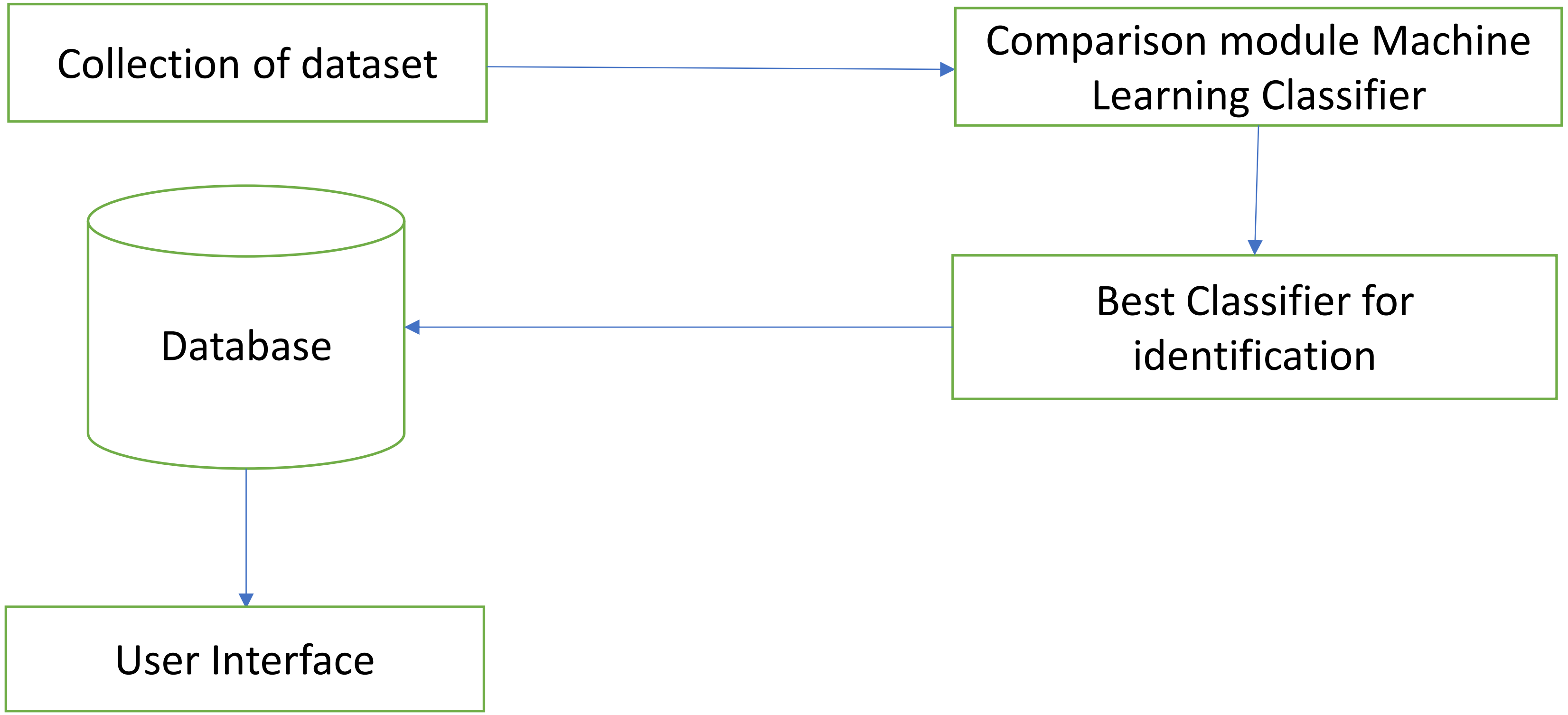
1. To design and implement machine learning algorithm to recommend which crop to grow based on soil pH and temperature.

2. Performance analysis of the two algorithms using machine learning.

3. To design and develop the user interface.

4. Methodology

BLOCK DIAGRAM :



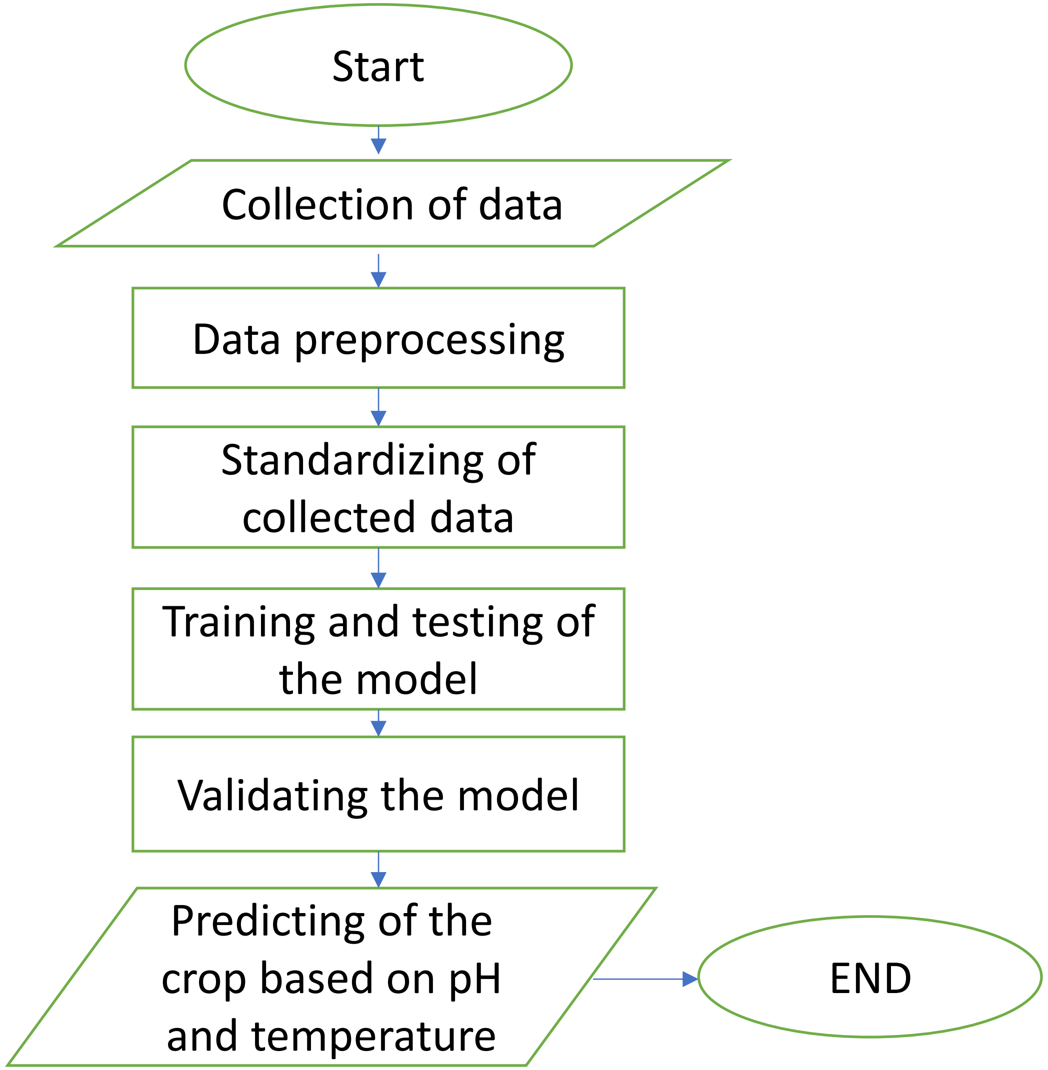
4. Methodology

In this project , we have used two algorithms – SVM (SUPPORT VECTOR MACHINE) and LSTM(LONG SHORT TERM MEMORY).Here we are going to compare accuracy of both these algorithms .Then will choose the algorithm which is of high accuracy.

A Support Vector Machine (SVM) is a powerful machine learning algorithm widely used for both linear and nonlinear classification, as well as regression and outlier detection tasks. SVMs are highly adaptable, making them suitable for various applications such as text classification, image classification, spam detection, handwriting identification, gene expression analysis, face detection, and anomaly detection.

Long Short-Term Memory is an improved version of recurrent neural network designed by Hochreiter & Schmidhuber. LSTMs can also be used in combination with other neural network architectures, such as Convolutional Neural Networks (CNNs) for image and video analysis.

5. Flow Chart


Start
Collection of data
Data preprocessing
Standardizing of collected data
Training and testing of the model
Validating the model
Predicting of the crop based on pH and temperature
END

6. Expected Results

1. Predicting the suitable crop to grow based on Ph and temperature.

2. Comparing the performance of two algorithms.

3. Taking inputs from user and predicting the crop.

7. Conclusion

1. The crop recommendation system ultimately helps farmers make informed decisions about which crops are best suited for their land, based on factors like soil type, temperature, and rainfall.
2. By using data-driven insights, the system can increase agricultural productivity, reduce risk, and promote sustainable farming practices, benefiting both farmers and the environment.
3. In addition to improving crop yield and resource utilization, the crop recommendation system empowers farmers with valuable knowledge, enabling them to adapt to changing environmental conditions and market demands.

THANK YOU